Michael Hall

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SUMMARY

Gameplay Engineer with 1+ years of professional experience building scalable AI and gameplay systems in Unreal Engine 5 for a live multiplayer shooter. Specialized in C++ gameplay architecture, AI behavior (Behavior Trees, EQS, Smart Objects), and network-safe, server-authoritative systems. Proven ability to refactor prototype logic into maintainable production code, collaborate with designers, and debug complex systemic and behavioral issues. Seeking a gameplay-focused engineering role where systems design, performance, and player experience intersect.

TECHNICAL SKILLS

- **Languages:** C++, C, Blueprint, Lua (basic), Scala (basic), Python (basic)
- **Game Development:** Unreal Engine 5, UE Gameplay Systems (gameplay ability system, multiplayer networking), AI (Behavior Trees, EQS, Smart Objects), Multiplayer Replication
- **Tools:** Git, Perforce, Wwise

EDUCATION

B.S. Computer Science (Software Engineering) — Bellevue University, *Class of 2026*, *Graduated with honors*

PROFESSIONAL EXPERIENCE

Contract Gameplay Engineer - Lost Boys Inc.:
"Project Midnight" / "No Man's Land" (Unreal Engine 5) (05/2026–Present)

- Engineered a soft lock-on camera system paired with motion warping for seamless, combat-responsive camera behavior, delivered within a 2-month investor-demo production window.
- Overhauled core combat and AI systems supporting the demo's combat loop as the engineer responsible for combat feel and AI behavior.
- Designed and implemented two enemy archetypes for the demo's encounters — a standard "Feral Soldier" grunt and a "Werewolf" mini-boss — each with distinct AI behavior and combat logic using C++, UE Blueprints, and Behavior Trees.

Freelance Gameplay Engineer - Studio Revive:
"Alter: Online" (Unreal Engine 5, Gameplay Ability System) (2026–Present)

- Architecting an adaptable morph and stat progression system using the Gameplay Ability System, mapping player-fed resources to artist-defined skeletal morph targets.
- Sole engineer on a companion-creature demo feature enabling dynamic, combinatorial appearance changes driven entirely by GAS attributes and abilities.

Associate Gameplay Engineer - Distinct Possibility Studios:
"Reaper Actual" (Shipped Title : Unreal Engine 5) (04/2024-12/2025)

- Designed and prototyped a unified Smart Object framework consolidating patrol and guard behaviors into a spline-driven system, establishing scalable architecture and designer-facing extensibility for future AI refactors.
- Implemented a fully replicable Auto Turret AI with server-authoritative targeting, quaternion-based rotation logic, and modular base classes; integrated Wwise SFX and animation controls for production readiness.
- Architected a global AI Ticketing Subsystem enabling dynamic coordination and behavior gating across NPCs, including server-side validation, async cleanup, and Behavior Tree integration.
- Refactored performance-critical AI logic by migrating Blueprint systems to native C++, optimizing EQS cover evaluation and reducing runtime overhead.
- Diagnosed and resolved replication, aiming, and synchronization defects using custom debug tools, runtime visualization, and trace-based validation.

PERSONAL & ACADEMIC PROJECTS

Tooth & Nail - Solo Developer, Unreal Engine

- Developing a fully networked 3D multiplayer fighting game showcasing readable directional combat, a server authoritative network model, and unique characters with real martial art inspired movesets.

AI-Assisted Imagination - Solo Developer, Unreal Engine

- Built a UE5 plugin integrating LLM-driven NPC dialogue with tunable personality and behavioral data assets.

References Available Upon Request